

Statistics for ML: Estimators, Sufficiency, Information

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The estimation vocabulary the MLE experiments keep touching, collected in one place. `statistics_demo.py` verifies the quantitative claims.

1 Estimators and their quality

An *estimator* is any function $\hat{\theta} = T(x_1, \dots, x_n)$ of the sample, targeting a parameter θ . Its basic report card:

$$\text{bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta, \quad \text{Var}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \mathbb{E}\hat{\theta})^2], \quad (1)$$

combined by the decomposition (expand the square around $\mathbb{E}\hat{\theta}$; the cross term vanishes):

$$\boxed{\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \text{bias}^2 + \text{Var}.} \quad (2)$$

Consistency: $\hat{\theta}_n \rightarrow \theta$ in probability as $n \rightarrow \infty$ — the LLN in estimator's clothing.

Remark 1 (Unbiasedness is a choice, not a virtue). *The Gaussian variance MLE $\frac{1}{n} \sum (x_i - \bar{x})^2$ has $\mathbb{E} = \frac{n-1}{n} \sigma^2$ (biased); Bessel's $\frac{1}{n-1}$ fixes the bias — yet the script measures the biased estimator to have the lower MSE at small n . Trading a little bias for less variance is often the better deal (2); regularization is this trade, industrialized.*

2 Sufficiency

Definition 1. $T(x_{1:n})$ is sufficient for θ if the conditional distribution of the sample given T does not depend on θ — once you know T , the rest of the data carries no further information about θ .

Theorem 1 (Fisher–Neyman factorization). T is sufficient for θ iff the likelihood factors as

$$L(\theta; x_{1:n}) = g(T(x_{1:n}), \theta) h(x_{1:n}), \quad (3)$$

with h free of θ .

Read off our worked examples: Bernoulli $L = p^m(1-p)^{n-m}$ — take $T = m$, $h \equiv 1$; Gaussian — expanding the exponent leaves θ multiplying only $\sum x_i$ and $\sum x_i^2$, so that pair is sufficient. This is why both MLEs are formulas in one or two summaries of the data rather than the data itself. (Rao–Blackwell sharpens the point: conditioning any unbiased estimator on a sufficient statistic never increases its variance — good estimators factor through T .)

3 Fisher information and the Cramér–Rao bound

Definition 2. The score is $\partial_\theta \log f(x; \theta)$; it has mean zero. The Fisher information per sample is its variance,

$$I(\theta) = \mathbb{E} \left[(\partial_\theta \log f(x; \theta))^2 \right] = -\mathbb{E} [\partial_\theta^2 \log f(x; \theta)], \quad (4)$$

the second form under regularity. For n i.i.d. samples information adds: $I_n = nI(\theta)$.

Intuition: I measures how sharply the log-likelihood peaks — how loudly one sample speaks about θ . Bernoulli: $I(p) = \frac{1}{p(1-p)}$ (extreme p are *more* informative per sample). Gaussian mean: $I(\mu) = 1/\sigma^2$.

Theorem 2 (Cramér–Rao). For any unbiased estimator $\hat{\theta}$ under regularity conditions,

$$\boxed{\text{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}}. \quad (5)$$

An estimator achieving (5) is *efficient*. The Bernoulli MLE $\hat{p} = \bar{x}$ is exactly efficient: $\text{Var}(\bar{x}) = p(1-p)/n = 1/(nI(p))$ — the script’s second panel shows the empirical variance sitting on the bound at every n .

4 Asymptotics of the MLE

Under regularity, the MLE is consistent and asymptotically normal:

$$\sqrt{n} (\hat{\theta}_{\text{MLE}} - \theta) \xrightarrow{d} \mathcal{N}(0, I(\theta)^{-1}), \quad (6)$$

i.e. the MLE attains the Cramér–Rao limit *asymptotically* even when no finite- n efficient estimator exists — the deep reason maximum likelihood is the default: nothing does better in the limit. Two practical corollaries: standard errors $\approx \sqrt{1/(n\hat{I})}$, and the invariance property (the MLE of $g(\theta)$ is $g(\hat{\theta})$) used by every reparameterization in our experiments.